

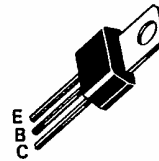
**NOT RECOMMENDED
FOR NEW DESIGNS**

UNIWATT SILICON ANNUULAR TRANSISTORS

Designed for complementary symmetry audio circuits to 10 Watt output.

- Low Collector-Emitter Saturation Voltage —
 $V_{CE(sat)} = 0.5 \text{ Vdc (Max) @ } I_C = 1.0 \text{ Adc}$
- Complements to PNP MPS-U51 and MPS-U51A
- Uniwatt Package for Excellent Thermal Properties —
1.0 Watt @ $T_A = 25^\circ\text{C}$

NPN SILICON AUDIO TRANSISTORS



MAXIMUM RATINGS

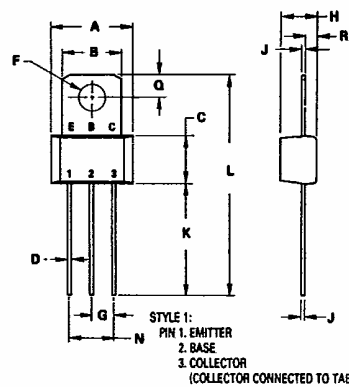
Rating	Symbol	MPS-U01	MPS-U01A	Unit
Collector-Emitter Voltage	V_{CEO}	30	40	Vdc
Collector-Base Voltage	V_{CB}	40	50	Vdc
Emitter-Base Voltage	V_{EB}	5.0		Vdc
Collector Current — Continuous	I_C	2.0		Adc
Total Power Dissipation @ $T_A = 25^\circ\text{C}$ Derate above 25°C	P_D	1.0	8.0	Watt mW/ $^\circ\text{C}$
Total Power Dissipation @ $T_C = 25^\circ\text{C}$ Derate above 25°C	P_D	10	80	Watts mW/ $^\circ\text{C}$
Operating and Storage Junction Temperature Range	T_J, T_{stg}	-55 to +150		$^\circ\text{C}$

THERMAL CHARACTERISTICS

Characteristic	Symbol	Max	Unit
Thermal Resistance, Junction to Case	$R_{\theta JC}$	12.5	$^\circ\text{C/W}$
Thermal Resistance, Junction to Ambient	$R_{\theta JA(1)}$	125	$^\circ\text{C/W}$

(1) $R_{\theta JA}$ is measured with the device soldered into a typical printed circuit board.

Uniwatt packages can be To-5 lead formed by adding -5 to the device title and tab formed for flush mounting by adding -1 to the device title.



NOTE:
1. LEADS WITHIN 0.15 mm (0.006) TOTAL OF TRUE POSITION AT CASE, AT MAXIMUM MATERIAL CONDITION.

DIM	MILLIMETERS		INCHES	
	MIN	MAX	MIN	MAX
A	3.14	9.53	0.360	0.375
B	6.50	7.24	0.260	0.285
C	5.41	5.66	0.213	0.223
D	0.36	0.53	0.015	0.021
F	3.18	3.33	0.125	0.131
G	2.54 BSC		0.100 BSC	
H	3.94	4.19	0.155	0.165
J	0.36	0.41	0.014	0.016
K	11.63	12.70	0.458	0.500
L	24.58	25.53	0.968	1.006
M	5.08 BSC		0.200 BSC	
O	2.39	2.69	0.094	0.106
R	1.14	1.40	0.045	0.055

CASE 152-02

ELECTRICAL CHARACTERISTICS (T_A = 25°C unless otherwise noted)

Characteristic		Symbol	Min	Max	Unit
OFF CHARACTERISTICS					
Collector-Emitter Breakdown Voltage (1) (I _C = 10 mA, I _B = 0)	MPS-U01 MPS-U01A	V _{(BR)CEO}	30 40	—	V _{dc}
Collector-Base Breakdown Voltage (I _C = 100 μA, I _E = 0)	MPS-U01 MPS-U01A	V _{(BR)CBO}	40 50	—	V _{dc}
Emitter-Base Breakdown Voltage (I _E = 100 μA, I _C = 0)		V _{(BR)EBO}	5.0	—	V _{dc}
Collector Cutoff Current (V _{CB} = 30 V, I _E = 0) (V _{CB} = 40 V, I _E = 0)	MPS-U01 MPS-U01A	I _{CBO}	— —	0.1 0.1	μA _{dc}
Emitter Cutoff Current (V _{BE} = 3.0 V, I _C = 0)		I _{EBO}	—	0.1	μA _{dc}
ON CHARACTERISTICS(1)					
DC Current Gain (I _C = 10 mA, V _{CE} = 1.0 V) (I _C = 100 mA, V _{CE} = 1.0 V) (I _C = 1.0 A, V _{CE} = 1.0 V)		h _{FE}	55 60 50	— — —	—
Collector-Emitter Saturation Voltage (I _C = 1.0 A, I _B = 0.1 A)		V _{CE(sat)}	—	0.5	V _{dc}
Base-Emitter On Voltage (I _C = 1.0 A, V _{CE} = 1.0 V)		V _{BE(on)}	—	1.2	V _{dc}
DYNAMIC CHARACTERISTICS					
Current-Gain-Bandwidth Product (I _C = 50 mA, V _{CE} = 10 V, f = 20 MHz)		f _T	50	—	MHz
Output Capacitance (V _{CB} = 10 V, I _E = 0, f = 1.0 MHz)		C _{ob}	—	20	pF

(1) Pulse Test: Pulse Width ≤ 300 μs, Duty Cycle ≤ 2.0%.

FIGURE 1 – DC CURRENT GAIN

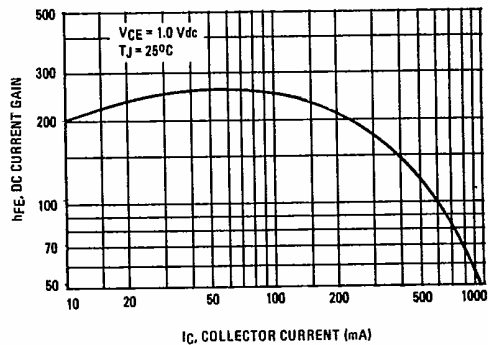


FIGURE 2 – "ON" VOLTAGES

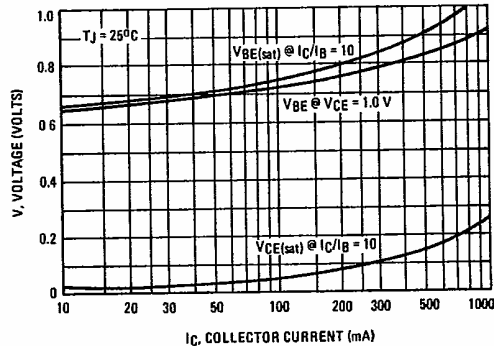
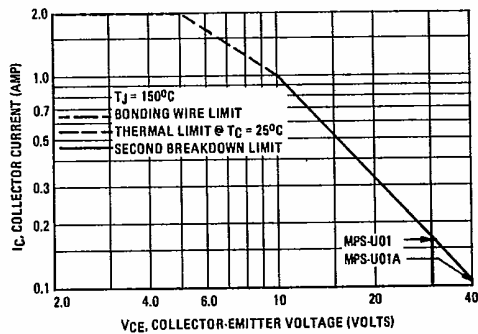


FIGURE 3 – DC SAFE OPERATING AREA



There are two limitations on the power handling ability of a transistor: junction temperature and secondary breakdown. Safe operating area curves indicate I_C–V_{CE} limits of the transistor that must be observed for reliable operation; i.e., the transistor must not be subjected to greater dissipation than the curves indicate.

The data of Figure 3 is based on T_{J(pk)} = 150°C; T_C is variable depending on conditions. At high case temperatures, thermal limitations will reduce the power that can be handled to values less than the limitations imposed by secondary breakdown.